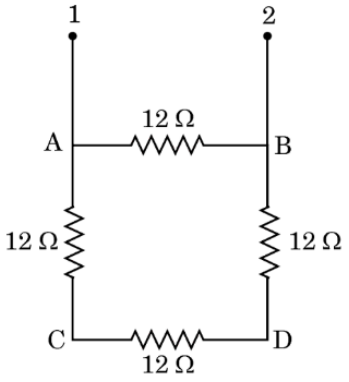
1) The correct way to connect an ammeter and a voltmeter in an electric circuit is :

(A) Ammeter in parallel and voltmeter in series
(B) Ammeter and voltmeter both in parallel
(C) Ammeter in series and voltmeter in parallel
(D) Ammeter and voltmeter both in series

2) An electric bulb is connected to a power supply of 220 V. If the current drawn by the bulb from the supply is 500 mA, the power of the bulb is :

(A) 11 W (B) 110 W
(C) 220 W (D) 1100 W

3) Four identical resistors of 12 Ω each are connected in series to form a square ABCD as shown in the figure. The resistance of the network between the two points 1 and 2 is :



(A) 48 Ω (B) 36 Ω
(C) 9 Ω (D) 6 Ω

4) A wire of length 'l' is gradually stretched so that its length increases to $3l$. If its original resistance is R, then its new resistance will be :

(A) 3R (B) 6R
(C) 9R (D) 27R

5) A piece of wire of resistance 'R' is cut lengthwise into three identical parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R', then the value of R/R' is:

(a) 1/9 (b) 1/3 (c) 3 (d) 9

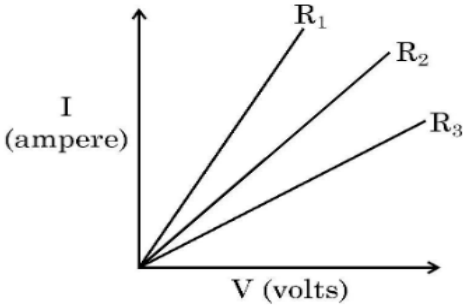
6) An electric bulb is rated 220 V; 11W. The resistance of its filament when it glows with a power supply of 220 V is:

(a) 4400 Ω (b) 440 Ω (c) 400 Ω (d) 20 Ω

7) The minimum number of identical bulbs of rating 4V; 6W, that can work safely with desired brightness, when connected in series with a 240 V mains supply is:

(a) 20 (b) 40 (c) 60 (d) 80

8) Study the I-V graph for three resistors of resistances R_1, R_2 and R_3 and select the correct statement from the following :



(A) $R_1 = R_2 = R_3$ (B) $R_1 > R_2 > R_3$

(C) $R_3 > R_2 > R_1$ (D) $R_2 > R_3 > R_1$

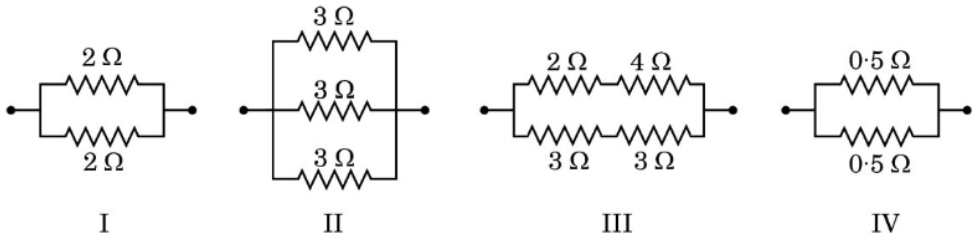
9) The maximum resistance of a network of five identical resistors of 1/5 Ω each can be:

(a) 1 Ω (b) 0.5 Ω (c) 0.25 Ω (d) 0.1 Ω

10) In case of four wires of same material, the resistance will be minimum if the diameter and length of the wire respectively are:

(a) D/2 and L/4 (b) D/4 and 4L (c) 2D and L (d) 4D and 2L

11) Consider the following combinations of resistors :



The combinations having equivalent resistance 1 Ω is/are :

(A) I and IV (B) Only IV

(C) I and II (D) I, II and III

12) An electric iron of resistance 20 Ω draws a current of 5 A. The heat developed in the iron in 30 seconds is:

(a) 15000 J (b) 6000 J (c) 1500 J (d) 3000 J

13) *Assertion (A) :* Electrons move from lower potential to higher potential in a conductor.

Reason (R) : A dry cell maintains electric potential difference across the ends of a conductor.

14) *Assertion (A) :* Nichrome is an alloy which is commonly used in electrical heating devices such as electric irons, toasters, etc.

Reason (R) : The resistivity of nichrome is high and its resistance decreases with increase in temperature.

15) **Assertion (A) :** In the common domestic circuits the earth wire is connected to a metallic plate buried deep inside the earth.

Reason (R) : Earth wire ensures that any leakage of current to the metallic body of the appliance keeps its potential to that of the earth, so the user may not get a severe electric shock.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is ***not*** the correct explanation of the Assertion (A).

(C) Assertion (A) is true, but Reason (R) is false.

(D) Assertion (A) is false, but Reason (R) is true.

16) At the time of short circuit, the electric current in the circuit:

(a) varies continuously (b) does not change (c) reduces substantially (d) increases heavily

17) If four identical resistors, of resistance 8 ohm, are first connected in series so as to give an effective resistance R_s, and then connected in parallel so as to give an effective resistance R_p, then the ratio $\frac{R_s}{R_p}$ is

(a) 32 (b) 2

(c) 0.5 (d) 16

18) In domestic electric circuits the wiring with 15 A current rating is for the electric devices which have

(a) higher power ratings such as geyser. (b) lower power ratings such as fan.

(c) metallic bodies and low power ratings. (d) non-metallic bodies and low power ratings.

19) The expressions that relate (i) Q, I and t and (ii) Q, V and W respectively are (Here the symbols have their usual meanings) :

(a) (i) $I = \frac{Q}{t}$ (ii) $W = \frac{V}{Q}$

(b) (i) $Q = I \times t$ (ii) $W = V \times Q$

(c) (i) $Q = \frac{I}{t}$ (ii) $V = \frac{W}{Q}$

(d) (i) $I = \frac{Q}{t}$ (ii) $Q = \frac{V}{W}$

20) An electric kettle consumes 1 kW of electric power when operated at 220 V. The minimum rating of the fuse wire to be used for it is:

(a) 1 A (b) 2 A (c) 4 A (d) 5 A

21) A current of 0.5 A is drawn by a filament of an electric bulb for 10 minutes. Find the amount of electric charge that flows through the circuit.

22) Calculate the number of electrons passing through a cross-section of a conductor in 1 second if a current of 1 A flows through it.

23) A potential difference of 6 V is applied across an electrical component, and a current of 0.5 A flows through it. Find the resistance of the component.

24) An electric bulb is rated 220 V, 100 W. Calculate the resistance of its filament.

25) A wire of resistance R is cut into 5 equal parts. These parts are then connected in parallel. If the equivalent resistance of this combination is R', find the ratio R/R'.

26) Calculate the heat produced in joules when a current of 2 A flows through a resistor of $4\ \Omega$ for 10 seconds.

27) Define current and its Unit.

28) Define voltage and its Unit.

29) State ohm's law & write its expression

30) Define Electric Power and write is unit.

31) State joules law of heating.

32) How much is 1 commercial unit of energy i.e. 1 UNIT is equal to SI unit of energy i.e. joules.

33) If you doubled a wire's length *and* doubled its thickness (cross-section), what happens to its resistance?

34) Why does a bulb glow but its connecting wire doesn't, though the same current flows through both?

35) If one fairy light fuses and the whole string goes dark, what does that say about how they're connected?

36) If electrons flow from negative to positive, why do we say current flows the opposite way?

37) You pay the electricity bill. What exactly are you paying for, if not electrons?

38) Two bulbs on the same 220 V line draw very different currents. What does that tell you about them?

39) Why does a "kilowatt hour" measure energy and not power, despite having "watt" in its name?

40) If the potential difference across a resistor is halved, what happens to the heat produced per second?